

Swesan Pathmanathan

Canadian Citizen

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EDUCATION

McMaster University

Bachelor of Engineering, Software Engineering (Co-op)

Hamilton, ON

Sept 2021 – Apr 2026

TECHNICAL SKILLS

Languages: C, C++, Python, Verilog

Systems: Linux, Bash, Windows, Quartus, Docker, ROCm, BMC, BIOS/UEFI

Validation: post-silicon GPU, RTL testbenches, Memory, IO, firmware interfaces, RAS, Performance, System Management

WORK EXPERIENCE

Advanced Micro Devices (AMD) | Python, C/C++, Verilog, Quartus, testbenches, post-silicon GPU, Memory, IO, firmware, Linux, BMC, BIOS/UEFI

Markham, ON

Datacenter GPU System Integration Engineer (Co-op)

May 2024 – Aug 2025

- Acted as rotating **test lead** for MI300X, MI325X, and MI350X **post-silicon** datacenter **GPU** validation lanes, orchestrating **100+ runs weekly** on **Linux** and **Windows** while HPC and ML stress workloads ran and companion test modules exercised **RAS, IO, Performance, Memory, and System Management** under validated baselines.
- Ran post-silicon validation focused on **Memory** and **IO** coverage on MI300X, MI325X, and MI350X datacenter **GPUs** beside HPC and ML workloads, surfacing interface-level regressions before wider integration gates.
- Debugged **BMC** and **BIOS/UEFI** issues on MI300X, MI325X, and MI350X systems from different customer environments, flashed and verified firmware as needed, and wrote team-facing BMC, BIOS/UEFI, and flash bring-up guides teammates reused on **20+ rack bring-ups, cutting repeat setup clarifications by about 40%** once pitfalls were centralized.
- Extended **Python** automation and an internal workload launcher with version-tracked scripts, shrinking diagnostic setup from **about 20 minutes to under 5 minutes** per run because engineers reused the same canned launch paths.
- Triaged defects across vendor **GPU** stacks including ROCm, firmware, **C/C++**, and interconnects using logs; added **repeatable test scripts** so every handoff shipped the same repro bundle, which **cut reopened and duplicate defects on my queue by about 30%** by eliminating one-off retests.

PROJECTS

ALU Design with Sequential Circuits | Verilog, Quartus

Sept 2025

- Structured a synthesizable ALU datapath from small **Verilog** blocks and checked behavior block-by-block before full-chip stitching.
- Coded deterministic **Verilog testbench** stimulus plus self-checks, exercised ALU arithmetic and logic modes over a scripted input matrix, and used **Quartus** waveform debug until every planned functional regression passed.

SufSort Sorting System | C, Bash, Linux

Nov 2025

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.
- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.

Breast Cancer Prediction Neural Network | Python, PyTorch, NumPy, scikit-learn

Sept 2025 – Dec 2025

- Built a feedforward neural network with **PyTorch**, using **NumPy** and **scikit-learn** preprocessing and stratified splits, to classify malignant and benign breast cancer samples from tabular clinical datasets.
- Applied regularization techniques including **dropout** and **L1/L2** penalties to improve model stability and reduce overfitting.

RELEVANT COURSES

Computer Architecture, Digital Systems Interfacing, Computer Performance Analysis, Software Testing, Real-Time Systems